

Algebra I Team Exam

2026 James Clemens Math Tournament

INSTRUCTIONS:

1. Your team has 60 minutes to complete this examination.
2. One official team from each school consisting of four students and other unofficial teams may compete. Only the scores from the official team will be counted towards a school's final score.
3. This exam consists of 15 free-response questions, each worth 10 points if answered correctly and 0 points if answered incorrectly or left unanswered.
4. Calculators, cellular devices, books, notes, and other aids are prohibited during this exam. Scratch paper will be provided for calculations. If you require additional scratch paper, simply raise your hand and your proctor will assist you.
5. Write your answers to the questions in the provided team answer sheet. You may use the test booklets for scratch work, but only the answers marked in the team answer sheet will be counted. Please write **legibly**.
6. **All answers to the team exam will be a positive integer from 1 to 999, inclusive.**
7. **Leave out** all units and special characters when recording your answer on the team answer sheet. For example, if you get \$1 as your answer, write 1 on the answer sheet.
8. A school's final score for a grade will be calculated by adding the top 4 written test scores in that grade (excluding tiebreaker questions) as well as the team round score of the official team, for a maximum of 550 points.
9. In the event of a tie, the school with the highest team round score will be favored. If team round scores result in another tie, answers will be evaluated starting backwards from question 15 to 1 to determine a winner.
10. Diagrams are not necessarily drawn to scale.
11. Although this math tournament is intended to demonstrate your knowledge and skills in math, be sure to enjoy the experience and have fun!
12. You may keep the test booklet after the exam.

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1. If $x + y = 20$ and $xy = 28$, then what is $x^2 + y^2$?
2. A circle of radius $\frac{5}{\sqrt{\pi}}$ is drawn within a square of area 100. What percent of the square lies within the circle?
3. There are 5 people sitting at a table: Person A, Person B, Person C, Person D, and Person E. Person A and B cannot sit next to each other. Given this restriction, how many ways can the people be seated around the table? (Two arrangements are considered identical if they are rotations of each other).
4. The median of 3 numbers a, b, c is twice their mean. If $a < b, c$ and their mode is 4, what is $|a + b + c|$?
5. What is $x + y + z$, given the following system of equations?
$$\begin{aligned}10x - 2y - z &= -4 \\8x + 2y - 5z &= 4 \\3x + 2y - 5z &= -6\end{aligned}$$
6. How many integers between 1 and 999 exist such that every digit of the integer is odd?
7. The roots of the quadratic $x^2 + mx + n$ are twice the roots of the quadratic $x^2 + px + m$, where $m, n, p \neq 0$. What is $\frac{n}{p}$?
8. The year 2026 can be written as a product of prime numbers. What is the sum of the exponents of all of these numbers? (Ex. $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 5^2 \cdot 2^2$ Add $2 + 2$ [the exponents] to get 4.)

9. Two cities, A and B, are 300 miles apart. Car 1 leaves A for B at 60 mph. One hour later, Car 2 leaves city B for A at 40 mph. At what distance from city A do they meet, in miles?
10. Two positive integers have a greatest common factor (GCF) of 4 and a least common multiple (LCM) of 60. If their sum is 28, what is the product of the two numbers?
11. There exists a number n for which $n!$ is divisible by all factors of 2026. What is the remainder when n is divided by 10?
12. A triangle has sides in the ratio of 3 : 5 : 7. The length of its shortest side is 12 cm. If the triangle is scaled down so that its shortest side becomes 6 cm, what is the length of the longest side in the scaled-down triangle?
13. A loaded six-sided die has a 1 in n chance of rolling a 1, a 2 in n chance of rolling a 2, a 3 in n chance of rolling a 3, and so on. If you roll 3 of these loaded dice, the probability that the sum of the three numbers is largest is $\frac{p}{q}$, where p and q are relatively prime positive integers. What is $p + q$?
14. What is the measure of one exterior angle of a regular icosagon (20 sided polygon)?
15. Sarah and Jackson are playing a game with n stones. Sarah goes first, with the players alternating turns. A player removes 1 or 3 stones from the pile on their turn. If the pile is empty when a player's turn begins, that player loses. What is the largest n , ($1 \leq n \leq 999$) for which Jackson wins?